

Conic Sections

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **classify** a circle, ellipse, or hyperbola from its equation or defining geometric relationship.

RELATED SKILLS

- I Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
- I Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.

- ☐ 2. I can **use** completing the square to rewrite a general-form conic equation in standard form.

RELATED SKILLS

- I Complete squares to convert a general conic equation to standard form.
- D Create a perfect-square trinomial by completing the square.
- A Simplify an algebraic expression using operation properties.

- ☐ 3. I can **write, interpret, and** graph implicit and parametric equations of a circle.

RELATED SKILLS

- I Connect implicit, standard, and parametric equations of a circle.
- I Determine and use a circle's center and radius.
- A Generate ordered pairs from parametric equations.
- A Graph a parametric curve with its direction of motion.

- ☐ 4. I can **write, interpret, and** graph implicit and parametric equations of an ellipse.

RELATED SKILLS

- I Connect implicit, standard, and parametric equations of an ellipse.
- I Determine and use an ellipse's center, axes, vertices, and foci.
- A Generate ordered pairs from parametric equations.
- A Graph a parametric curve with its direction of motion.

- ☐ 5. I can **write, interpret, and** graph implicit and parametric equations of a hyperbola.

RELATED SKILLS

- I Connect implicit, standard, and parametric equations of a hyperbola.
- I Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
- A Generate ordered pairs from parametric equations.
- A Graph a parametric curve with its direction of motion.

- ☐ **6. I can **determine** the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.**

RELATED SKILLS

- ☐ **D** Determine and use a circle's center and radius.
 - ☐ **D** Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes.
 - ☐ **D** Determine and use an ellipse's center, axes, vertices, and foci.
 - ☐ **A** Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition.
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- ☐ **7. I can **explain** the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.**

RELATED SKILLS

- ☐ **D** Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas.
 - ☐ **A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
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- ☐ **8. I can **explain** a parabola as the set of points equidistant from a focus and directrix.**

RELATED SKILLS

- ☐ **I** Explain a parabola using its focus-directrix distance relationship.
 - ☐ **A** Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
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